

Thomas Tompion and English watchmaking

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Thomas Tompion was born in 1639 in Bedfordshire. Both his father and his grandfather were blacksmiths. At that time, most clocks were made of iron for churches by blacksmiths. These clocks were not very accurate and needed to be reset from a sundial quite frequently. Typically, they only had hour hands.

The English clockmakers, most of whom were immigrants, obtained a charter for the Clockmakers Company in 1631.

The industry was transformed by the application of the pendulum to regulate a clock by Huygens in Holland in 1656. This changed the clock from an indifferent timekeeper to a fairly accurate timepiece. This development was successfully exploited by English makers from 1658.

Tompion probably started work as a blacksmith, and possibly made one or two iron clocks. In 1664 he became an apprentice to a London clockmaker. Little is known of his early career in London; the first reference is in 1670 when he was recorded in Water Lane (now Whitefriars Street) off Fleet Street.

Tompion became known as the “Father of English watchmaking” because of his important innovations.

In 1695 he took an apprentice, George Graham, and the address was recorded as the “Dial and 33 Crowns”, Corner Water Lane, Fleet Street. Graham too made some important innovations.

The first watches were nothing like the watches we know today – they were more like small portable clocks which could be carried round, often supported on a string round the neck. The growing fashion for waistcoats prompted a demand for a slimmer watch which could be slipped into the waistcoat pocket.

In 1658, Robert Hooke described a spring-regulated mechanism for watches, but he never got round to building one until a parallel invention by Huygens on the Continent prompted him to start a collaboration with Thomas Tompion.

They built the first balance-regulated spring-driven watch and presented it to King Charles II, inscribed “Hooke Inventit 1658 Tompion fecit 1675”. This improvement

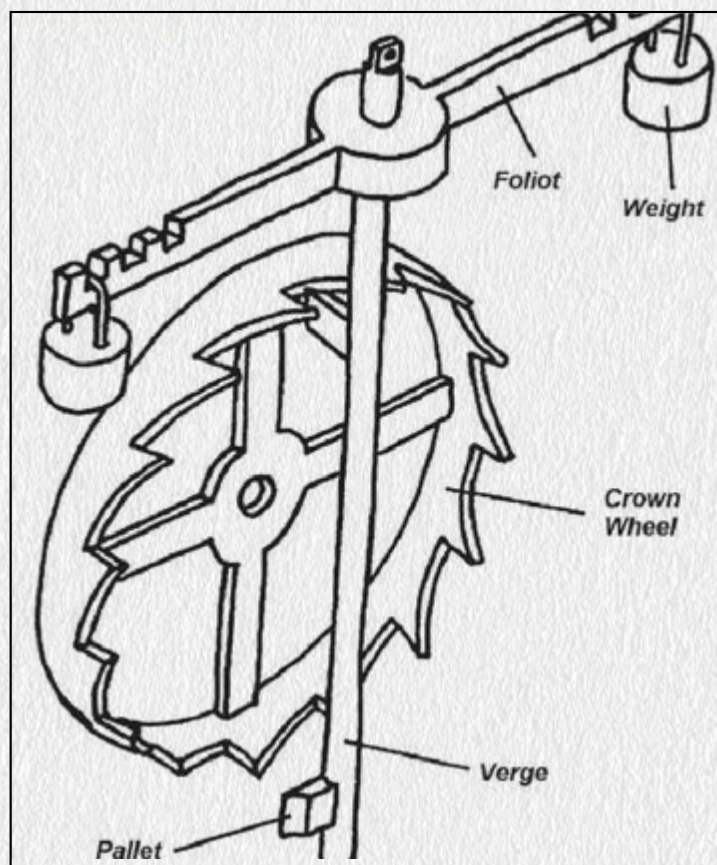
greatly increased the accuracy of the watch. Hooke also worked on improvements to pendulum clocks using an anchor escapement.

Tompion’s excellence and unrivalled reputation were based on the sound design of his products, the high quality of the materials used, and the outstanding skills of the workmen he employed, many of whom were of French or Dutch Huguenot origin. Sir William Petty highlighted the very early example of the division of labour and its effect on cost.

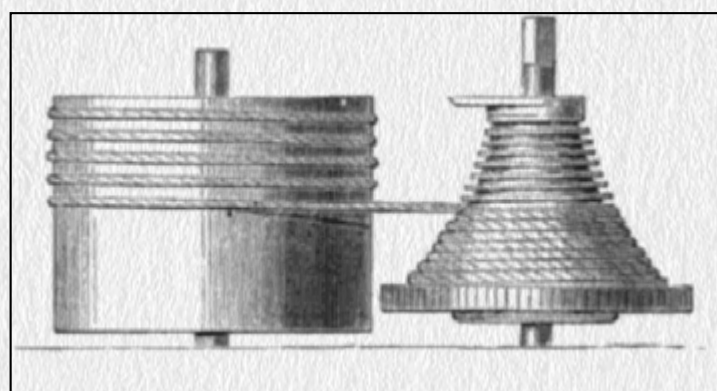
English clockmakers compensated for the declining power of the mainspring by using a cone-shaped spindle called the fusee. Although it introduced another bulky component into the watch, it greatly improved accuracy and added to the reputation of English-made watches.

The last twenty years of Tompion’s life included successive improvements to escapements to improve the accuracy of watches. The cylinder escapement introduced by George Graham in the 1720s built on all this work.

Thomas Tompion died in 1713, and George Graham continued the business, which moved in 1720 to the “Dial and 1 Crown” near Fleet Bridge. Both were Fellows of the Royal Society, and both are buried in Westminster Abbey.



Schematic of verge and foliot mechanism



Mainspring barrel and fusee